

Anjana Raman

December 6, 2025

DS 4002

Food Classifier Case Study Rubric

Why am I doing this? This is an opportunity for you to work with real data to explore how qualitative information such as ingredients lists can be turned into meaningful data used for classification. As a data scientist it's important to understand how design choices can greatly impact the way a model behaves and the outcomes that come out of it. This exercise mirrors the foundational work required before any predictive model can be built: determining how labels, rules, and heuristics are created in the first place.

What am I going to do? You will be given a cleaned dataset from the USDA that includes a list of branded foods and their ingredient lists. With the starter code you will be:

1. Reviewing ingredient groups and deciding if they are “negative” or “positive”
2. Assigning weights to various ingredient groups (how “healthy” or “unhealthy” these ingredients should be counted)
3. Determine if long ingredients should incur penalty, and how drastic this penalty should be
4. Set thresholds on what scores should be counted as healthy or unhealthy
5. Write a short report explaining your current scoring system, how it compares to the baseline, and your reasoning.

Tips for Success:

- Use the current provided ingredients as a baseline to work off of. Figure out how to weigh these ingredients before creating new categories (or changing them).
- Do some preliminary research on widely accepted healthy and unhealthy ingredients as the basis of your logic for scoring ingredients.
- Be consistent with weighting logic. If a precedent is set for a certain ingredient, similar ingredients should be treated similarly.
- Test your code early. If all foods come out as either unhealthy or healthy, you may need to tweak your logic.

Spec Category	Description
Formatting	Goal: Make your submission readable, professional, and easy for graders to interpret. The Report should be a PDF file with all necessary analysis and findings along with any code and outputs you created.
Report File	Goal: Demonstrate your understanding of ingredient-level scoring and your ability to justify your decisions. Your PDF must include: – A description of your positive and negative ingredient groups – The weights and penalties you assigned – Your thresholds for Healthy / Intermediate / Unhealthy – A comparison table showing how your labels matched or differed from the baseline – A thoughtful reflection that explains the strengths and weaknesses of your rubric and what surprised you
Code	Goal: Allow graders to reproduce your scoring results and follow your logic step-by-step. Your GitHub repo should include: – The provided starter notebook with your completed scoring functions – Any additional helper functions or modifications you made – Clear comments explaining your design choices Be sure to cite any outside references or inspirations used in forming your ingredient logic.